

Process Grammar Model

Immediate Grammar and Adjustive Grammar

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Contents

1	Introduction	3
2	Theoretical Background	3
2.1	Relation to Dual Process Theory	3
2.2	Limitation of “spoken language” and “written language”	3
2.3	Clause chaining and additive constructions	4
2.4	Clause chaining and linking methods of phrases	4
3	Immediate Grammar and Adjustive Grammar	5
3.1	Immediate Grammar	5
3.2	Adjustive Grammar	6
4	Concept Diagram	9
4.1	Temporal Dynamics of Processing Dominance	9
4.2	Overlap of Immediate Grammar and Adjustive Grammar	10
4.3	Continuum of Immediate Grammar and Adjustive Grammar	10
4.4	Adjustment Saturation Point	11
5	Tasuki-gake/cross-directional Effect	11
6	Description Method	12
6.1	Design of Appropriate Format	12
6.2	Consideration at the Meta-level of Distinction	12
6.3	Confirmation of Distinction by Empirical Data	13
7	Future Directions	13
7.1	Research Questions	13
7.2	Update Policy	13
8	Conclusion	13
A	About the Q&A Section	14
A.1	Q&A: Immediate Grammar and Its Theoretical Background	14
B	Considerations from Examples and Practice	14
C	Accumulation of Previous Research Supporting the Existence of Immediate Grammar	15
C.1	Accumulation of Observations Supporting the Existence of Immediate Grammar	15
C.2	The Conventional Concepts of “Spoken Language vs. Written Language” and the Position of PGM	16
C.3	The Three Layers: Grammar, Expression, and Effect	16
C.4	Wallace Chafe’s Information Structure Model and the Process Grammar Model (PGM)	16
C.5	Comparison of Halliday’s Functional Grammar and Generative Theory, and the Positioning of PGM	17

1 Introduction

Immediate grammar is a grammar that corresponds to the situation where utterances are intuitively selected and promptly realized. Adjustive grammar is a grammar used for utterances with appropriate judgment and adjustive. As a framework for describing language use with two extremes, one that must be spoken immediately and one that is spoken after careful consideration and adjustive, we propose a process grammar model. Immediate grammar is not just about speaking anything right away, but there are strict rules for this. Adjustive grammar is important to choose the right words and use the right grammar, but depending on the degree of adjustive (or the degree of revision), there are several ways of expression, and there is a state of over-adjustment. This model, which considers the time axis of expression, is not something completely different from all previous grammar studies, but a framework for further developing previous grammar studies.

The feature of this model is to describe the dynamic characteristics of grammar in language considering the time axis. Language is operated through physical resources (cognitive processes of the brain, speech, symbol manipulation, etc.), making it possible to describe it mathematically. However, language is not just a physical system, and each part that describes matter has different properties from the physical quantities described in physics, making mathematical descriptions difficult. Physical systems (thermodynamics, electromagnetism, etc.) usually follow deterministic laws. On the other hand, language contains non-deterministic elements such as “immediate grammar” and “adjustive grammar.” In particular, language includes context dependency, semantic structure, and cognitive process influences, which cannot be fully explained by simple mathematical models. Therefore, a mathematical description of language must consider dynamic changes and decision-making processes while being based on physical resources. For example, real-time generation of speech (immediate grammar) is difficult to predict completely with deterministic laws, so it should be relatively understood and always show pairs to be compared like a binary opposition. For example, semantic ambiguity (e.g., “She has a daughter” is it my daughter or a girl?) and context dependency (e.g., “She met him” is it she met him or he met her?) are difficult to resolve mathematically. These problems cannot be ignored, nor can they be simply addressed by statistical probability theory. However, a mathematical model of language that considers the time axis may offer a new approach to these unresolved issues. For example, if it is immediate grammar, considering the real-time generation of speech, the problem of my daughter or a girl is resolved by the relationship before and after the speech. Also, if it falls under adjustive grammar, considering context dependency, the problem of ambiguity between she met him and he met her is resolved by adding words during the adjustment process.

2 Theoretical Background

Please refer to **Q1.** as well.

2.1 Relation to Dual Process Theory

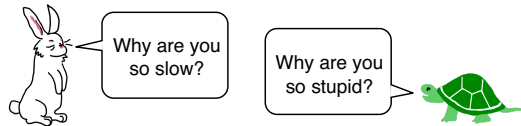


FIGURE 1: Dual processing theory based on the presentation from “System 1 and System 2 Thinking”

The Process Grammar Model is similar to the framework of “System 1 (fast thinking) and System 2 (slow thinking)” referred as the Dual Process Theory. However, this model has a unique perspective focusing on immediacy and adjustability, rather than being a mere application. (Evans 2008; Kahneman 2011; Squire and Kandel 2009)

2.2 Limitation of “spoken language” and “written language”

In traditional linguistics, it has been said that “spoken language has high immediacy and written language is revisable.” However, in SNS and live streaming, the situation of “speaking while writing” is increasing,

TABLE 1: Differences in characteristics between System 1 and System 2

System 1	Example utterance immediate	System 2	Example utterance deliberate
Fast	“Whoa, that’s hot!”	Slow	“Be careful, it’s hot enough to burn you.”
Subconscious	“I have a bad feeling about this...”	Conscious	“There are several risk factors here.”
Automatic	“Oh, go ahead!”	Effortful	“Should I say ‘please go ahead’...?”
Heuristic	“It’s always like this, so it’ll be the same this time.”	Analytical	“Looking at the past five cases, this one might be different.”
Associative	“This smell... reminds me of something.”	Rule-based	“This fits the typical pattern according to theory X.”
Implicit	“Yeah, kind of like that.”	Explicit	“A leads to B, and that causes C.”
Emotional	“I can’t take this anymore!”	Logical	“There are three problems with that approach.”
Intuitive	“Something just feels off.”	Rational	“Considering the costs and risks, this option is more reasonable.”
Error-prone	“It’ll probably be fine!”	Reliable	“I’ve verified it with multiple sources.”

and it cannot be explained by a simple binary opposition. This study proposes the name “Process Grammar Model” as a framework to explain the mechanism of language use, rather than calling surface acts as language forms, to organize such phenomena.

2.3 Clause chaining and additive constructions

“Immediate grammar” is most closely related to the linguistic phenomenon of clause chaining that has been studied to date. Clause chaining has been widely studied as a method to indicate grammatical continuity. By linking multiple clauses while maintaining the structure and meaning of sentences, clause chaining expresses the continuity and relationships of information. Clause chaining is a useful method for explaining the continuum of immediate and adjustive grammars.

Sakakura (1975) expressed the construction of Japanese text as “adding one after another,” and Komatsu (2003) stated that it is “constructed by adding one after another,” pointing out the characteristics of clause chaining in Japanese poetry and prose. Such expressions emphasize that the construction of sentences is sequential and flexible, suggesting that the relationships between each phrase are not necessarily close. Kondo (2005) also stated that “since it is added, it is difficult to define which is the main clause and which is the subordinate clause in the resulting compound sentence,” indicating that the hierarchical structure of clauses is ambiguous. These considerations suggest that language use is based on the natural order in which acts and events are recalled, rather than following syntactic binding rules. They reveal that language utterances are based on flexible adaptations to the flow of the speaker’s thoughts and the situation, rather than grammatical adjustments. Therefore, such immediate speech processes are considered to be universally observed features in language development and communication.

2.4 Clause chaining and linking methods of phrases

Clancy (2020) studied the use of clause chaining in Japanese narratives, where 60 children (aged 3-7) and 10 adults retold stories after viewing picture book storytelling and short videos. This study examined the methods of linking phrases, the timing of their completion, and semantic relationships (simultaneity, causality, temporal order) in detail. The timing of phrase completion was confirmed to be related to changes in characters and the end of important story units. Furthermore, the results of comparing children’s and adults’ stories showed that factors influencing the linking of phrases exist regardless of age.

The responses in children’s stories are thought to naturally appear in a form close to immediate grammar. When telling stories, children often repeat responses in short units, and it is noted that the chaining of phrases is not always logically coherent. This is considered to reflect the characteristic of immediate grammar, where sentences are constructed immediately. Children’s immediate responses are dominant at a stage where linguistic adjustments are minimal, and they intuitively connect to the next phrase. This immediate response plays an important role in the process of language development and serves as a foundation for learning complex syntax and grammatical adjustments. The flexibility and

fluidity that children exhibit when creating sentences are considered to be related to the characteristics of immediate grammar.

3 Immediate Grammar and Adjustive Grammar

TABLE 2: Comparison of Immediate Grammar and Adjustive Grammar

	Immediate Grammar	Adjustive Grammar
Nouns, noun phrases only	Just now, curry udon, done. So to speak, like music notation.	The curry udon has just been completed. This corresponds to music notation.
Omitting “wa” particle	How about my explanation?	Is my explanation being conveyed properly?
Adverbs only	Certainly.	Certainly, that point is valid.
Demonstratives only	This becomes like this.	This element changes like this.

Immediate grammar is short and has high real-time characteristics. It is established in the smallest unit. Traditionally, it has not been considered as a sentence, but such forms are more prominent and widely used in SNS and live streaming. Adjustive grammar has a clear sentence structure and is a grammar that can be revised.

To confirm the existence of immediate grammar, it is generally necessary to shorten what is commonly called a sentence and capture the form when it can be spoken briefly to the person in front of you, classifying it with names such as “noun stop” and “adverbs only” to reveal its characteristic rules. Here, to succinctly show the characteristics of immediate grammar and adjustive grammar, the comparison of the two is shown in Table 2.

3.1 Immediate Grammar

Immediate grammar is formalized as a dynamic process of generating language in real-time. This model is described as a system that determines appropriate output immediately based on input (situation and context).

Characteristics: Instantaneous speech, frequent omissions, context-dependent, real-time processing

Examples: “How’s the taste?,” “Look!,” “Watch out!,” “Ordinary is the best”

Application: Everyday conversation, interjections, emergency speech

3.1.1 Basic Model

In the Process Grammar Model, Immediate Grammar can be represented by the following function.

$$f_{IG} : (C, M, T) \rightarrow O$$

where C represents context, including external conditions such as the relationship between speaker and listener, the setting of the utterance, and visual information, as well as internal conditions related to the emergence of language processing, such as recollection or awareness. M represents linguistic knowledge, including vocabulary, grammatical structures, and grammatical knowledge available for immediate language generation. T represents the temporal conditions under which language processing unfolds. O represents the output, namely the generated linguistic expression, including actual utterances and written linguistic expressions. Immediate Grammar is a process in which language processing emerges reactively in response to an external or internal change under the conditions (C, M, T) , generating an output O appropriate to the situation at that moment.

3.1.2 Condition of Immediacy

The core of Immediate Grammar lies in the reactive emergence of language processing in response to an external or internal change. Here, “immediacy” does not mean that processing must be completed within a fixed amount of time. Rather, it refers to a processing characteristic in which linguistic expression begins to emerge reactively in the situation at hand. Thus, short processing time is a frequent characteristic of Immediate Grammar, but it is not treated as a defining condition.

3.1.3 Dynamic Adaptability

Immediate Grammar adaptively changes its output in response to changes in input. This dynamic adaptability can be represented as the following optimization problem.

$$\arg \max_O U(O \mid C, M, T)$$

where $U(O \mid C, M, T)$ is a utility function representing the appropriateness of output O under context C , linguistic knowledge M , and temporal conditions T . The expression represents the selection of a more appropriate linguistic output O under the conditions obtaining at that moment.

3.1.4 Representation of Continuity

Immediate Grammar is viewed not as a process that produces fixed outputs discretely, but as a dynamic process in which output may vary in response to changes in context and temporal conditions. This sensitivity can be represented conceptually as follows.

$$\frac{\partial O}{\partial C} \neq 0, \quad \frac{\partial O}{\partial T} \neq 0$$

where $\partial O / \partial C$ represents the sensitivity of output O to changes in context C , and $\partial O / \partial T$ represents its sensitivity to changes in temporal conditions T . These expressions conceptually indicate that the output of Immediate Grammar is not fixed independently of contextual and temporal conditions. In Immediate Grammar, language processing emerges reactively in response to such changes without requiring conscious adjustment to precede it.

3.1.5 Overall Expression of Immediate Grammar

Integrating the above elements, Immediate Grammar can be expressed in the following form.

$$O = f_{IG}(C, M, T)$$

Here, the output O of Immediate Grammar is generated by the function f_{IG} on the basis of context and situation C , linguistic knowledge M , and temporal conditions T . The variable T does not represent a threshold that defines Immediate Grammar as processing completed within a fixed amount of time; rather, it represents the temporal conditions under which language processing unfolds. The essential property of Immediate Grammar is that language processing emerges reactively in response to an external or internal change, generating an output O appropriate to the situation at that moment.

3.2 Adjustive Grammar

Characteristics: Careful selection, grammatically correct structure, revision

Examples: “Based on the results of this study...,” “Thank you very much”

Application: Formal speeches, papers, official documents

Adjustive grammar differs from immediate grammar in that there is a conscious adjustment of grammatical forms. The most typical example is formal grammar. It is less likely that subjects and complements are omitted. To limit the content or object, adjectives and adverbs are often placed before nouns. However, in actual sentences, the order of phrases is not absolutely fixed.

3.2.1 Definition of Adjustive Grammar

Adjustive grammar is a process that generates language expressions that are carefully adjusted based on context and purpose, planned with temporal flexibility. To design adjustive grammar, it is necessary to clearly define what adjustment means. Adjustment is understood as the process of adapting and modifying the meaning or form of utterances. For example, it refers to clarifying utterances, correcting information, and aligning intentions with others, including elements such as “clarification of meaning,” “correction of misunderstandings,” “grammatical corrections,” and “maintenance of communication flow.” However, when making adjustments such as adding content or persuasion, it is important to be cautious of the Adjustment Saturation Point (to be discussed later).

3.2.2 Basic Structure of Adjustive Grammar

The basic structure of adjustive grammar is designed along the following flow: First, regarding the initial utterance, it provides information in the initial utterance (ending with a predicate, providing basic facts, etc.). As the start of adjustment, if there is a change in recognition regarding the meaning or structure intended by the initial utterance, it indicates the beginning of that correction. Adjustive grammar clearly records this timing of “correction” and indicates what has been corrected. Regarding the process of adjustment (reconstruction), if correction is necessary, it records how the adjustment was made (grammatical reconstruction, addition of information, reordering of utterances, etc.). After adjustment, the final utterance (corrected utterance) indicates what kind of semantic consistency was maintained as a result of the completed utterance after adjustment. By concretely formatting this flow, it becomes clearer how adjustive grammar functions.

3.2.3 Theoretical Background of the Adjustment Process

To theoretically support the format of adjustive grammar, it is necessary to demonstrate what cognitive and social processes adjustment involves. For example, it is important to reflect in language that adjustment is a process that “reduces uncertainty in communication” and that it is an “adaptive correction based on the assumption of the listener’s understanding.” As a cognitive element, adjustment is a cognitive process that accurately conveys the meaning of utterances to the listener or resolves misunderstandings, making the process dynamic, with the act of adjustment itself being instantaneous. As a social element, adjustment is also a social adaptation in conversation and a collaborative effort with the listener, involving the alignment and confirmation of intentions between interlocutors. As long as these are related to language adjustment, adjustive grammar functions as a format to reflect cognitive and social processes in language expression. However, when it comes to content adjustment, there is a possibility of reaching the Adjustment Saturation Point from that moment, so care must be taken in the design of adjustive grammar.

3.2.4 Feedback Loop in Adjustive Grammar

In adjustive grammar, it is also possible to incorporate the concept of a feedback loop. This reflects the process of receiving feedback after utterance and making corrections based on that content. This allows us to show that adjustment is not just a one-time correction but a dynamically evolving process. For example, in a series of flows such as utterance → understanding → re-confirmation → adjustment, we record changes or corrections to utterances at each stage.

3.2.5 Specific Examples of Adjustive Grammar

For example, while there is a simple rule in news scripts that ends with a predicate, in adjustive grammar, there are cases where additional information or corrections are made afterward. In this case, the following structure can be considered in adjustive grammar: The initial utterance states the fact that “an earthquake has occurred.” At the start of adjustment, additional information such as “the epicenter is in eastern Japan” is added. The process of adjustment records the clarification of meaning through the addition of information. The adjusted utterance generated afterward is “An earthquake has occurred. The epicenter is in eastern Japan, and the magnitude is predicted to be 6.5.”

3.2.6 Rule Design for Adjustive Grammar

As for modification rules, it is important to define the timing and methods for making adjustments (such as adding information, rephrasing, or confirmation questions). Types of changes include semantic corrections, changes in word order, and modifications to grammatical structures. Thus, the format of adjustive grammar is crucial for clearly expressing the process of “adaptive correction” based on theoretical backgrounds, and it is important to uniformly describe specific methods for each stage and the dynamic changes in that process.

3.2.7 Basic Model

Adjustive grammar is defined by the following function.

$$f_{AG} : (C, M, P, T) \rightarrow O$$

Here, C represents context (situation and background), which includes topics, the listener's background knowledge, and social relationships. M refers to linguistic knowledge (vocabulary, grammatical structures), including syntax and expression choices specific to written language. P signifies planning (purpose and adjustment strategies), encompassing the intention of utterances (persuasion, explanation) and style (formal, casual). T represents temporal flexibility, indicating that the larger it is, the wider the scope for adjustment. Finally, O signifies output (generated language expression), including refined utterances or polished texts.

3.2.8 Condition of Planning

In adjustive grammar, the utterance plan is an important element. The specificity of the plan is expressed by the following function.

$$P = g(C, M)$$

Here, g is a function that generates plans based on context C and linguistic knowledge M . The generation of plan P enables the production of adjusted language expressions.

3.2.9 Temporal Flexibility

In Adjustive Grammar, temporal flexibility T does not function as a threshold that determines the presence of adjustive processing. Rather, it is a factor that affects the extent to which planning, selection, and revision can be applied to linguistic expression. Greater temporal flexibility allows a wider range of adjustive operations, such as lexical selection, structural organization, and clarification of meaning. However, an increase in available time does not in itself indicate a transition to Adjustive Grammar; the relative processing dominance of Immediate Grammar and Adjustive Grammar varies according to context, communicative goals, and processing demands.

3.2.10 Optimization of Adjustment

In adjustive grammar, output O is optimized based on plan P . This optimization can be expressed by the following equation.

$$\arg \max_O U(O \mid C, M, P, T)$$

$U(O \mid C, M, P, T)$: Utility function representing the appropriateness or achievement of purpose of output O .

3.2.11 Representation of Continuity

In adjustive grammar, continuous adjustments are made according to context and plan. This continuity is shown by the following differential equations.

$$\frac{\partial O}{\partial C}, \frac{\partial O}{\partial P} \neq 0$$

where, $\frac{\partial O}{\partial C}$ represents the sensitivity of adjustment to changes in context, and $\frac{\partial O}{\partial P}$ represents the sensitivity of adjustment to changes in plan.

3.2.12 Overall Expression of Adjustive Grammar

The overall expression of Adjustive Grammar is described as follows.

$$O = f_{AG}(C, M, P, T)$$

Here, the output O of Adjustive Grammar is generated on the basis of context and situation C , linguistic knowledge M , planning P , and temporal conditions T . The variable T does not represent a threshold that determines a transition to Adjustive Grammar; rather, it affects the extent to which adjustive operations such as planning, selection, and revision can be carried out. The essential property of Adjustive Grammar lies not in the duration of processing itself, but in the adaptive modification of linguistic expression according to communicative goals and context.

3.2.13 Adjustment Saturation Point

In language use, speakers and writers adjust linguistic expressions through lexical selection, rephrasing, structural organization, and clarification of meaning according to communicative goals and context. However, such linguistic adjustment has a limit. The point at which further adjustment is no longer expected to improve the linguistic expression, and further change would require modification or evaluation of the content itself, is referred to as the **Adjustment Saturation Point**.

Let c_{adj} represent the degree of adjustment, and let c_{max} represent the upper limit within which the process can still be treated as linguistic adjustment. The applicable range of Adjustive Grammar can then be represented as follows.

$$c_{adj} \leq c_{max} \quad (1)$$

where c_{max} represents the Adjustment Saturation Point. Changes required beyond this point are treated not as adjustments of linguistic form but as modifications or evaluations of the content itself. Thus, the Adjustment Saturation Point is not the temporal endpoint of Adjustive Grammar, but the limit of what can be treated as linguistic adjustment.

3.2.14 Specific Examples and Interpretation of the Adjustment Saturation Point

At the Adjustment Saturation Point, the process reaches the limit of what can still be treated as linguistic adjustment, and the issue shifts from a problem of linguistic form to one concerning the content itself. The following example illustrates this transition.

Theater Script

A theater script is adjusted with consideration for whether the audience can understand the content, whether the lines are easy for actors to deliver, and whether the flow of the scene is natural. Lexical choice, rephrasing, and the organization of sentence length and structure fall within the range of Adjustive Grammar as adjustments of linguistic form. However, once the issue becomes whether a particular meaning should be expressed explicitly in dialogue or conveyed through silence, gesture, facial expression, or stage direction, the problem is no longer one of linguistic adjustment but of dramatic composition and performance. This transition can be regarded as an example of the Adjustment Saturation Point.

3.2.15 Significance of the Adjustment Saturation Point

The Adjustment Saturation Point clarifies the range of Adjustive Grammar by limiting it to the adaptation and modification of linguistic form. Changes or evaluations required beyond this point are treated not as linguistic adjustment but as separate issues concerning the content itself. Thus, the Adjustment Saturation Point defines not only the applicable range of Adjustive Grammar but also the outer boundary of the linguistic domain described by the Process Grammar Model.

Thus, Immediate Grammar concerns the reactive emergence of language processing in response to external or internal change, whereas Adjustive Grammar concerns the adaptation, modification, and confirmation of linguistic expressions. Although the two have distinct processing characteristics, they do not operate as mutually exclusive systems in actual language use, but function through interaction with one another.

4 Concept Diagram

Here, we present a conceptual diagram showing how the relative processing dominance of Immediate Grammar and Adjustive Grammar changes over time (Figure 2).

4.1 Temporal Dynamics of Processing Dominance

IG and AG do not exist as two distinct intervals that clearly alternate on the time axis. At the emergence of language processing, IG tends to dominate reactive processing in response to external or internal change, while over time AG becomes more dominant in planning, integration, and revision. In this model, this change in relative processing dominance is conceptualized as the temporal dynamics of language processing.

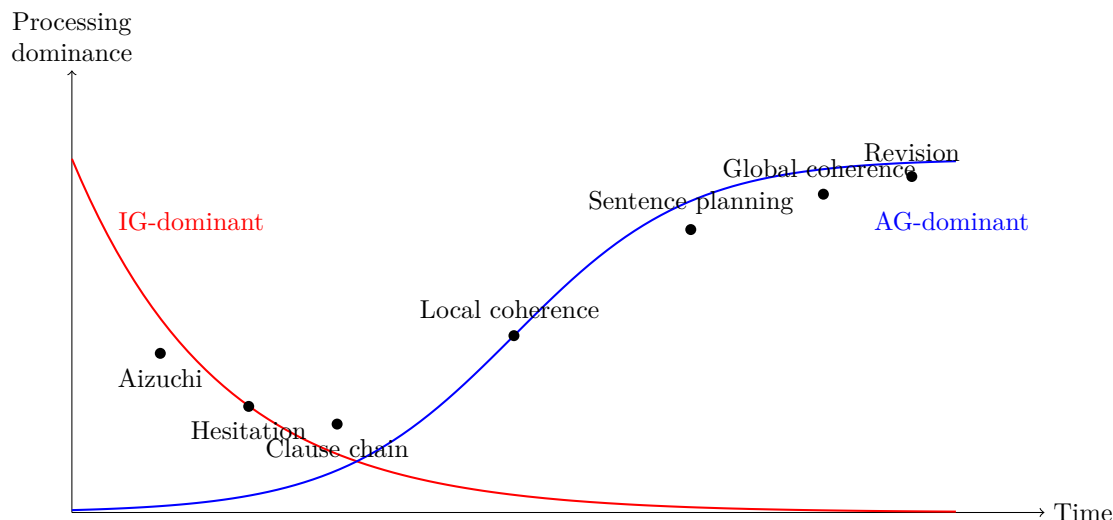


FIGURE 2: Temporal dynamics of processing dominance in the Process Grammar Model. The horizontal axis represents time in the unfolding of language processing, and the vertical axis represents relative processing dominance. The red curve conceptually represents the tendency for the processing dominance of Immediate Grammar (IG) to decrease over time, while the blue curve represents the tendency for the processing dominance of Adjustive Grammar (AG) to increase. Black dots indicate approximate positions of typical linguistic phenomena. The rabbit and turtle function as visual metaphors for IG and AG processing.

4.2 Overlap of Immediate Grammar and Adjustive Grammar

A decrease in IG dominance does not imply the disappearance of IG. Similarly, the onset of AG involvement does not imply the cessation of IG. In actual language processing, there are periods in which both operate simultaneously, while their relative processing dominance changes over time. This coexistence is conceptualized as overlap in this model.

4.3 Continuum of Immediate Grammar and Adjustive Grammar

The relationship between immediate grammar and adjustive grammar is not a simple binary opposition, but can be regarded as a "continuum" in the sense that their involvement varies depending on context, goals, and experience. In this study, the continuum does not mean that a third grammar exists between them. Rather, it refers to the fact that Immediate Grammar and Adjustive Grammar may overlap in time within language processing, while their relative processing dominance may change continuously. For example, in language learning, beginners rely heavily on adjustive grammar (System 2) when constructing sentences, but through training, frequently used patterns may become automatized, enabling immediate output (the use of System 1). Through such skill acquisition, the division of labor between System 1 and System 2 can change even for the same task. Accordingly, operations once handled by adjustive grammar may later be incorporated into immediate generation. This variability constitutes the core of the "continuum" in the Process Grammar Model, demonstrating the complementary and inseparable relationship between immediate grammar and adjustive grammar.

However, treating Immediate Grammar and Adjustive Grammar as a continuum does not mean that they are identical processing modes. Immediate Grammar focuses on the reactive emergence of language processing, whereas Adjustive Grammar focuses on the adaptation and modification of linguistic expression. They have distinct processing characteristics, while overlapping and changing in relative dominance within actual language processing.

This shift in processing dominance also corresponds to the relationship between System 1 and System 2 in dual process theory. Fast processing is often associated with System 1, but this does not necessarily mean that such processing is innately System-1-based (Kahneman 2011; Evans 2008). Through experience and repetition, operations that were initially carried out by System 2 may become automatized and executed as fast processing. (Evans 2008) Even for problems that would normally require careful delib-

eration, training and repeated practice can lead to automatized processing that appears System-1-like. In language learning, sentence construction that was not immediately available at an early stage may become immediately usable after mastery. On the other hand, when explaining new words or complex procedures, careful System 2 adjustment is likely to be required in order to avoid misunderstanding. Thus, even within the same linguistic task, the relative processing dominance of IG and AG may vary according to experience and context.

4.4 Adjustment Saturation Point

The Adjustment Saturation Point refers to the point at which further adjustment of a linguistic expression is no longer expected to improve the expression. Beyond this point, the task is no longer one of linguistic adjustment, such as lexical selection, rephrasing, structural reorganization, or clarification, but instead requires modification or evaluation of the content itself. Thus, the Adjustment Saturation Point is not the temporal endpoint of Adjustive Grammar, but the limit of what can still be treated as linguistic adjustment.

For example, in legal writing, refining definitions and making conditions more explicit in order to reduce ambiguity falls within the domain of Adjustive Grammar. However, once the issue becomes whether the provision itself is substantively appropriate, the problem is no longer one of linguistic form but of legal content. This transition provides a typical example of the Adjustment Saturation Point.

5 Tasuki-gake/cross-directional Effect

The Tasuki-gake Effect does not represent an exception to the distinction between Immediate Grammar and Adjustive Grammar. Rather, it demonstrates that utterance dynamics, linguistic form, and perceived effect belong to distinct layers. Even an utterance produced immediately may employ an Adjustive Expression that has become formulaic or automatized through practice, while an Immediate Expression may be intentionally used in a carefully planned or revised text. Moreover, this crossed relationship is not merely accidental; speakers and writers may exploit it intentionally to produce stylistic effects. This chapter treats such crossed correspondences as the Tasuki-gake Effect.

More specifically, the **Tasuki-gake Effect** refers to a stylistic effect arising from a crossed correspondence between the grammatical processing underlying an utterance, the linguistic form realized through that processing, and the effect produced by the expression. When an Adjustive Expression is used in an immediate utterance, it may produce effects such as politeness or formality. Conversely, when an Immediate Expression is employed in planned or revised language use, it may create naturalness, immediacy, or a conversational impression. In this model, such stylistic effects produced through a crossed relationship between processing, expression, and effect are referred to as the Tasuki-gake Effect.

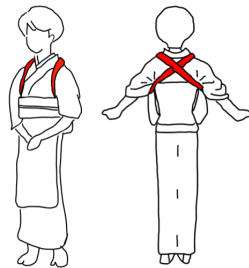


FIGURE 3: Tasuki-gake: a method of tying a long cloth strap around the back to pull the sleeves of a kimono together, so that they cross over the back. This is done to prevent the sleeves from getting in the way while working.

In this model, utterance is treated as a dynamic phenomenon unfolding over time. An Immediate Utterance is an utterance in which Immediate Grammar is relatively dominant, whereas an Adjustive Utterance is one in which Adjustive Grammar is relatively dominant. These are not two temporally discrete or mutually exclusive types of utterance; in actual language processing, they may overlap and appear with changing relative processing dominance.

Immediate Grammar refers to the grammatical principles that support the reactive emergence of language processing in response to external or internal change, whereas Adjustive Grammar refers to the

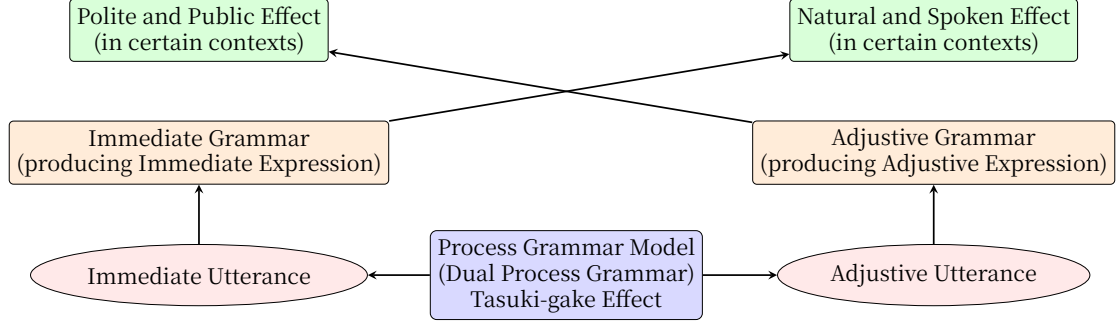


FIGURE 4: Tasuki-gake Effect: a visual representation of the crossed relationship among processing, expression, and effect in the Process Grammar Model. Like the crossed structure of a traditional tasuki sash, the diagram illustrates how the grammatical origin of an expression and the effect produced by its realization may cross.

grammatical principles that support planning, selection, revision, and adaptation of linguistic expression. The linguistic forms realized through these processes are referred to as Immediate Expression and Adjustive Expression, respectively.

A crucial point in this model is that the grammatical process underlying an utterance, the linguistic form realized through that process, and the effect perceived by a listener or reader do not necessarily correspond on a one-to-one basis.

For example, when an Adjustive Expression becomes automatized through practice or formulaic use and is employed in immediate speech, it may produce a Polite and Public Effect, creating an impression of politeness, formality, or public appropriateness. Conversely, when an Immediate Expression is intentionally incorporated into a novel, interview article, script, or other planned medium, it may produce a Natural and Spoken Effect, creating an impression of naturalness, conversationality, or immediacy.

Thus, the relationship between the grammatical origin of an expression and the effect produced by its realization is not fixed. The two may form a crossed relationship. In the Process Grammar Model, this crossed correspondence is referred to as the Tasuki-gake Effect (Figures 3 and 4).

6 Description Method

6.1 Design of Appropriate Format

When describing the relationship between Immediate Grammar and Adjustive Grammar, a format is required that preserves their distinct processing characteristics while representing their overlap and changes in relative processing dominance. For Immediate Grammar, the format should represent the reactive emergence of language processing in response to external or internal change. For Adjustive Grammar, it should represent processes such as planning, selection, revision, and confirmation of linguistic expression.

These processes should not be represented as two mutually exclusive intervals on the time axis. Rather, the format should allow them to overlap within the same course of language processing while their relative processing dominance changes. Such a description makes it possible to preserve the distinct processing characteristics of Immediate Grammar and Adjustive Grammar while representing their complementary relationship.

6.2 Consideration at the Meta-level of Distinction

Immediate Grammar and Adjustive Grammar may overlap in actual language processing, but they have distinct processing characteristics. Accordingly, their distinction should be examined not merely in terms of temporal sequence, but in terms of which type of processing is relatively central.

Immediate Grammar is characterized by the reactive emergence of language processing in response to external or internal change, whereas Adjustive Grammar is characterized by adaptive processes such as planning, selection, revision, and confirmation of linguistic expression. The distinction between the two can therefore be understood in terms of these differences in processing characteristics.

6.3 Confirmation of Distinction by Empirical Data

To provide empirical support for the theory, it is necessary to examine how the processing characteristics of Immediate Grammar and Adjustive Grammar appear in actual conversation and text production. Individual cases should not be classified exclusively as either Immediate Grammar or Adjustive Grammar. Rather, processing characteristics such as reactive emergence, planning, revision, and confirmation should be observed in order to describe their relative processing dominance and degree of overlap.

By positioning actual linguistic data along multiple dimensions, it becomes possible to visualize how the characteristics of Immediate Grammar and Adjustive Grammar are distributed and how their relative processing dominance changes continuously. Such empirical analysis can be used to examine the validity of overlap and continuum in the Process Grammar Model.

One possible descriptive approach is to represent, within the same linguistic data, the reactive processing characteristic of Immediate Grammar and the adjustment processes characteristic of Adjustive Grammar, and to compare their relative involvement. By describing processing dominance and the degree of overlap on the basis of actual data, the parameters of a continuum model in which immediacy and adjustability interact can be examined empirically.

TABLE 3: Characteristics of Immediate Grammar and Adjustive Grammar

Item	Immediate Grammar	Adjustive Grammar
Characteristics	Grammar that supports the reactive emergence of language processing in response to external or internal change.	Grammar that supports adaptive processes such as planning, selection, revision, and confirmation of linguistic expression.
Typical Examples	Reactive responses, backchannels, and expressions generated in response to the situation at hand.	Rephrasing, lexical reselection, structural reorganization, and revised expressions.
Processing Characteristics	Reactive processing is relatively dominant.	Processing involving the adjustment of linguistic form is relatively dominant.
Primary Function	To generate linguistic expressions appropriate to the situation at that moment.	To enhance the appropriateness, clarity, and accuracy of linguistic expression.

7 Future Directions

7.1 Research Questions

- Formalization of relative processing dominance, overlap, and the continuum between Immediate Grammar and Adjustive Grammar
- Collection and analysis of empirical data on processing characteristics in conversation, text production, and other forms of language use

7.2 Update Policy

- Refinement of descriptive rules and formalization for Immediate Grammar and Adjustive Grammar
- Empirical examination and refinement of the models of processing dominance, overlap, and continuum
- Application to language education, AI, and natural language processing
- Extension to diachronic and stylistic analyses of waka, narratives, and historical texts

8 Conclusion

This study has presented the basic framework of the Process Grammar Model and examined the definitions of Immediate Grammar and Adjustive Grammar, the temporal dynamics of their processing dominance, their overlap and continuum, the Adjustment Saturation Point, and the Tasuki-gake Effect.

Rather than treating linguistic form solely as a fixed system, the model provides a dynamic framework for describing how language processing emerges, is adjusted over time, and is realized as linguistic expression. Future work will further refine the descriptive framework and test the model against empirical data, while exploring its applications to language education, AI, and the analysis of historical and stylistic materials.

A About the Q&A Section

In this book, we have presented the theoretical model of immediate grammar and explored a new framework for language understanding through examples and applications based on it. However, when introducing such a new theory, it is expected that various questions and points of confirmation will arise from the reader's perspective.

This "Q&A Section" aims to record the author's position and supplementary considerations regarding such questions. Additionally, this section will include questions and discoveries that have emerged during the translation work (for example, "Tosa Nikki" and "Ise Monogatari") and the daily collection of immediate grammar expressions (aead project).

To avoid "subtle discrepancies that arise every time I write" and to maintain the coherence of thought, this section will be used. It is assumed that the integration with the main text will be reconsidered later as needed, and this section is positioned as a flexible space for accumulation and examination.

A.1 Q&A: Immediate Grammar and Its Theoretical Background

Q1. What kind of theory is Immediate Grammar? How is it positioned in relation to existing syntactic theories?

A. Immediate Grammar focuses on the immediate and flexible forms of expression observed in language use and proposes them as structures that can be theoretically explained. This model is not positioned in opposition to syntactic-centered theories like phrase structure grammar but is a hypothetical framework that aims to describe human language behavior from a different perspective.

Q2. In what ways does Immediate Grammar contribute to language description?

A. By introducing the perspective of Immediate Grammar, utterances that have previously been considered "exceptional" or "lacking structure" can be positioned as a systematic form of language use. This can be seen as evidence that Immediate Grammar provides a theoretical "slot for explanation" for such utterances.

B Considerations from Examples and Practice

Q3. Is Immediate Grammar the same as spoken language?

A. Immediate Grammar is often confused with "spoken language," but theoretically, they are different concepts. Immediate Grammar is a grammatical model based on the timing of psychological processing and structural flexibility, and even in oral expressions, adjusted grammar can be used.

Q4. Why don't you use corpus-based automatic extraction?

A. In recent language research, methods for extracting syntactic patterns from corpora have been widely used. This is a very effective method for analyzing formal structures (word order, dependency, part-of-speech patterns). However, what Immediate Grammar targets is not just the form of syntax but the "way of speaking," "way of responding," and "expressions chosen at that moment," which cannot be judged solely by syntactic labels or structural information. For example, utterances like "そうそう、それぞれ" (yes, yes, that's it) or continuous expressions like "うん、でもさあ" (yeah, but you know) may seem grammatically ambiguous and fragmented, but they have a sophisticated structure based on immediate judgments, emotions, and the flow of responses. Such phenomena, even if present in the corpus, are often overlooked due to reasons like "not matching search criteria" or "not recognized as grammatical coherence," and it is believed that they can be more accurately handled through contextual judgment by human eyes and descriptive work based on experience. Therefore, in this study, we adopt an observational approach based on description and linguistic intuition, not in opposition to existing corpus-based methods but as a complementary one.

Q5. Does Immediate Grammar ignore the procedures of recording, transcription, and analysis like conversation analysis or discourse analysis?

A. No, the position of Immediate Grammar in this study shares many commonalities with the "specificity of utterances" and "contextual dependence" targeted by discourse analysis and conversation analysis. In particular, the continuity of utterances, such as word choice according to context, timing, backchanneling, and rephrasing, is also an important observation target of Immediate Grammar. However, the analytical methods differ. In conversation analysis and discourse analysis, the procedure of recording audio → transcribing → structural analysis of utterance units is used to describe the interaction between participants. In contrast, the purpose of Immediate Grammar is to describe the patterns observed in such utterances as a model of the structures that speakers instantaneously select and manipulate. Therefore, while this study does not adopt the procedures of recording and transcription, it sufficiently references utterance materials and examples based on them, taking a complementary approach in that the focus of observation is on the "immediate structures being used." In short, Immediate Grammar does not contradict the results of discourse analysis or conversation analysis; rather, it aims to function as a structural explanatory model underlying the "detailed utterance phenomena" they depict.

Q6. How are chaining constructions or chaining grammar related to Immediate Grammar?

A. Chaining constructions or chaining grammar represent a common form used when constructing complex sentences in a dynamic and intuitive way. They are a key manifestation of Immediate Grammar. For instance, sentences formed with expressions like "and then," "but," or "so," which are added successively as one speaks, exemplify the nature of Immediate Grammar, as they are generated without pre-planning the entire sentence. These forms closely correspond to how people construct language in real-time.

Q7. Why is it easier for non-native speakers to use constructions with "de" or "te" rather than complex noun phrases with "no"?

A. Japanese constructions using "no" require stacking modifiers in front of a noun, which demands structural planning and is characteristic of Adjustive Grammar. In contrast, constructions using "de" or "te" allow for sequential chaining of events or states, enabling spontaneous, step-by-step utterance—hallmarks of Immediate Grammar. For instance, the phrase "kinō eki de mita akai bōshi no onna no ko" ("the girl in the red hat I saw at the station yesterday") involves a complex noun phrase with multiple layers of modification. Meanwhile, a sentence like "Kinō eki ni itte, akai bōshi o kabutta onna no ko o mita" ("I went to the station yesterday and saw a girl wearing a red hat") follows a chronological flow and allows the speaker to build the sentence incrementally. This is similar in English: "The girl who wore a red hat that I saw at the station yesterday" is structurally demanding, while "I went to the station yesterday and saw a girl. She was wearing a red hat." is easier to produce and understand, especially for language learners.

C Accumulation of Previous Research Supporting the Existence of Immediate Grammar

C.1 Accumulation of Observations Supporting the Existence of Immediate Grammar

In this book, we present the theoretical framework of "Immediate Grammar," which is supported by multiple practical observations and records made by the author. For example, in the process of translating works like "Ise Monogatari" and "Tosa Nikki," many immediate expressions were found that are difficult to explain using conventional syntactic theories. Additionally, in the *aead* (An expression a day) project, we have been recording expressions with characteristics of immediate grammar in everyday Japanese, tagging them, and adding annotations to highlight the correspondence between language use and immediate grammar.

These tasks are not based on intuitive claims but are the results of observations accumulated over time. Hundreds of examples have been interpreted and described from the perspective of immediate grammar, providing empirical clues that demonstrate the effectiveness of immediate grammar as a descriptive model.

C.2 The Conventional Concepts of “Spoken Language vs. Written Language” and the Position of PGM

Traditionally, the classification of language styles has relied on a binary opposition or a continuum model between “spoken language (orality)” and “written language (literacy).” For example, Biber (1988) statistically analyzed the syntactic and lexical features of multiple text types and demonstrated that spoken and written styles exist across different media. Thus, the definitions of “spoken = spoken language” and “written = written language” have already been revised, leading to the proposal of more refined style models.

However, even in these models, the criteria for determining spoken vs. written language primarily depend on frequency of occurrence and lexical tendencies, and they do not adequately address the questions of “why that form is used” or “how that form is generated.” In particular, pre-prepared language, such as dialogue in fiction or speech scripts, can lead to inconsistencies between media and syntactic modes.

In this regard, the Process Grammar Model (PGM) redefines “spoken language” and “written language” not as attributes based on the medium of use, but as syntactic types based on the processing modes during language generation (immediacy vs. adjustability). That is, expressions generated by Immediate Grammar are spoken-like, while those based on Adjustive Grammar are written-like. This framework allows for the theoretical description of “spokenness” that is not spoken and “writtenness” that is not written.

C.3 The Three Layers: Grammar, Expression, and Effect

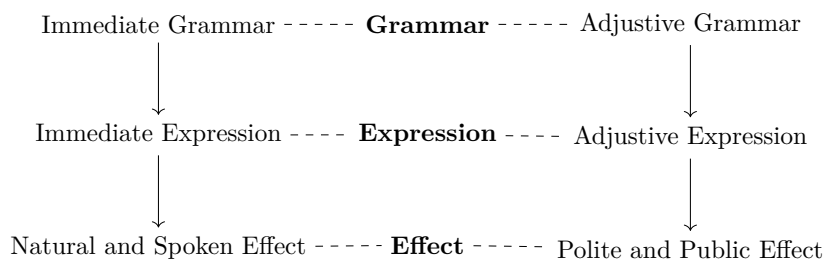


FIGURE 5: Three-layer structure in the Process Grammar Model (without intersection)

To describe language use more precisely, the Process Grammar Model adopts a three-layer structure: “Grammar,” “Expression,” and “Effect.” This structure focuses not on surface features but on the generation process, contextual use, and the interpretive effect on the listener or reader.

- **Grammar:** Immediate Grammar (real-time generation) vs. Adjustive Grammar (generated with preparation)
- **Expression:** The actual linguistic form expressed (Immediate vs. Adjustive Expression)
- **Effect:** Depending on the context, expressions evoke either a Natural and Spoken Effect or a Polite and Public Effect

This tripartite structure makes it possible to explain communicative effects even when the grammar and context do not align. For example, Immediate Expressions generated by Immediate Grammar, when used in novels or films, create a vivid and realistic impression (Natural and Spoken Effect). Conversely, Adjustive Expressions, when used in real-time speech such as public addresses, produce a refined and formal impression (Polite and Public Effect).

This cross-contextual usage is termed the “Tasuki-gake Effect,” highlighting how immediacy and adjustiveness can be flexibly combined beyond stylistic or media-based constraints.

C.4 Wallace Chafe’s Information Structure Model and the Process Grammar Model (PGM)

Chafe (1982) theoretically organized the differences between spoken and written language from the perspectives of “immediacy” and “adjustability,” proposing the following four conceptual axes:

- **Idea Units**
Speech is constructed by the sequential generation of “fragments of thought” as units of meaning, emotion, and rhythm. This forms the basis of the immediacy of spoken language.
- **Integration**
The degree to which information is compressed and structurally integrated. In written language, integration is enhanced through modification, nominalization, and complex sentence structures.
- **Involvement**
Indicates how much the speech is constructed based on the speaker’s emotions and relationships. The higher the involvement, the more immediate and empathetic the expression becomes.
- **Detachment**
An abstracted and objective descriptive style. When the narrator’s stance moves away from emotional involvement, the language takes on a detached quality, approaching adjustive grammar.

TABLE 4: Comparison of Chafe’s Model and PGM. Adapted from Chafe (1982)

Chafe’s Concept	Immediate Grammar (IG)	Adjustive Grammar (AG)
Idea Units	Spurt-like chain of speech	Integrated complex sentence structure
Integration	Low (including redundancy in word order and syntax)	High (embedding of modifiers, density of structure)
Involvement	High (emotional expression of the narrator, empathy)	Low (objectivity, depersonalized perspective)
Detachment	Low (specific involvement of the speaker)	High (abstraction, passivation, formality)

This comparison positions the two axes of grammatical use defined by PGM—“immediacy” and “adjustability”—within Chafe’s four concepts in psychological and discourse dimensions: “Idea Units” (fragmentation of thought), “Integration” (information density), “Involvement” (emotional and interpersonal connections), and “Detachment” (abstraction and depersonalization).

As shown in Table 4, immediate grammar is characterized by fragmented, emotional, and interpersonal language use, with a strong subjective involvement of the speaker. In contrast, adjustive grammar features highly integrated structures and moves toward abstract and objective forms of expression. Chafe’s theory plays an important role as an external theory supporting PGM’s “continuum model,” visualizing the complementary relationship between the two models.

However, Chafe’s model also has some limitations. First, his two axes of “integration” and “involvement” place emphasis on psychological and discourse perspectives, which may not sufficiently explain formal constraints in the language structure itself (e.g., word order, particles, structural permissibility of sentences). Second, while his theory of “idea units” is effective in demonstrating the fragmentation of spoken language, it has limited applicability in handling syntactic chains in narrative flow (e.g., anaphora, repetition, and vocative phrases).

In contrast, PGM explicitly describes immediate and adjustive grammar in terms of grammatical forms and generation conditions, providing a framework that adds structural dimensions to Chafe’s psychological dimensions. Thus, Chafe’s theory serves as an effective reference framework for broadening the descriptive depth of PGM, while PGM can be positioned to complement the structural and grammatical limitations of Chafe’s model.

Thus, while Chafe’s psychologically grounded model is highly effective in conceptualizing immediacy and adjustability, it remains limited when applied directly to grammatical description. The Process Grammar Model (PGM) inherits this theoretical legacy and extends it by integrating structural and generative dimensions of language.

C.5 Comparison of Halliday’s Functional Grammar and Generative Theory, and the Positioning of PGM

C.5.1 Halliday’s Basic Attitude Toward Spoken and Written Language and the Necessity of SFL

Halliday (Halliday and Hasan (1994)) views language not merely as a tool for communication but as a “social resource for creating meaning.” From this perspective, he particularly focuses on the differences

between spoken language and written language, placing the functional distinctions of each at the center of linguistic theory.

Halliday considers spoken language to be a linguistic activity rich in immediacy and involvement, characterized by contextual dependency, rhythm, and improvisation. In contrast, written language possesses a highly integrated structure, allowing for abstract and generalized expressions of meaning.

This functional distinction between spoken and written language led Halliday to develop Systemic Functional Linguistics (SFL). In SFL, language is not merely a collection of syntactic units but is viewed as a process that simultaneously fulfills three meta-functions: ideational (representing ideas), interpersonal (interacting with others), and textual (organizing text).

In particular, Halliday characterizes written language as “grammatically intricate” and spoken language as “lexically dense but structurally simple,” theorizing the differences in meaning functions that cannot be captured by traditional syntax-centered models. This serves as the starting point for his development of Systemic Functional Linguistics (SFL), providing a foundation for describing the choices of language that are contextually relevant to social situations.

C.5.2 Differences in the Purpose of Theoretical Construction

Halliday’s Systemic Functional Linguistics (SFL) and Chomsky’s Generative Theory (Chomsky (1965)) fundamentally differ in their basic understanding of language and the purpose of theoretical construction. Generative theory views language as an innate ability of humans, aiming to mathematically clarify its internal structure and syntactic rules. In this framework, an idealized speaker-listener is assumed, and sentences are primarily analyzed based on syntactic units such as NP (noun phrase) and VP (verb phrase). The focus is on constructing a universal grammar (UG) based on principles of economy and minimalism. In contrast, Halliday’s functional grammar views language as a resource for social action, aiming to clarify the meaning functions of language used in actual contexts. Here, the focus is not on an ideal speaker but on real writers and speakers, with clauses (not phrases) being the central unit of analysis. It is rooted in the actual use of meaning, emphasizing functional validity in context rather than syntactic correctness.

Thus, Chomsky focuses on the “internal structure” of grammar, while Halliday emphasizes the “social function” of grammar, with each theory attempting to address different questions. Each theory addresses different dimensions of linguistic questions and holds its own significance. Generative theory provides deep insights into the universal structure of language ability and its mathematical description, contributing significantly to the formal refinement of grammatical theory. On the other hand, SFL reflects the diversity of language use in social contexts and offers an important theoretical foundation in practical fields such as education, language acquisition, and media studies by viewing grammar as a means of meaning generation.

C.5.3 Halliday’s Strengths and Differences from Generative Theory

Halliday’s functional grammar possesses several strengths that differ from generative theory. First, Halliday approaches language from the perspective of use, making everyday grammatical usage—such as imperatives, desideratives, and evaluative expressions—the starting point of his theory. Second, he describes the structure of sentences not in terms of syntactic frameworks like subjects and predicates but in terms of semantic units such as actions, participants, and situations. Third, he assumes that language use always occurs within a social context, treating it as inseparable from social conditions such as style (register) and genre. Fourth, he emphasizes that a single sentence can simultaneously realize three meta-functions: ideational (representing ideas), interpersonal (interacting with others), and textual (organizing text). This layered structure is what makes SFL theoretically distinctive and demonstrates the integration of multifunctionality that has not been adequately addressed by other grammatical theories.

Critiques from the generative side have been directed at SFL (Newmeyer 1998; Pullum 2010). First, the system of rules in SFL lacks formally rigorous grammatical generation rules, which is criticized as a “lack of formal definition.” Second, while the relationship between syntax and meaning in SFL is close, it is seen as incompatible with the generative principle of “distinguishing syntax from semantics,” leading to perceptions of ambiguity. Third, SFL’s structural descriptions are considered insufficient in terms of mathematical consistency and reproducibility, making it difficult to model them as formal grammars.

This way, the flexible and meaning-oriented framework of SFL and the clarity and formal rigor sought by generative theory are both important for elucidating linguistic facts, and it is believed that an integrated framework should leverage these strengths.

C.5.4 PGM's Contribution to the Description of Social Functions in Grammar

The Process Grammar Model (PGM) functions as a supplementary framework that retains the social function perspective emphasized by SFL while making it more explicitly connected to grammatical structures. In particular, PGM focuses on the “conditions of use” under which linguistic expressions emerge, paying attention to whether speech is generated immediately or based on pre-adjusted syntax. This axis of “immediacy/adjustability” is deeply connected to Halliday’s interpersonal and textual functions, allowing for the description of the factors behind syntactic choices in actual language use.

Additionally, the concept of the “Effect Layer” introduced by PGM serves as an effective means to describe not only how syntactic forms carry meaning but also what pragmatic effects they produce (e.g., familiarity, respect, emphasis, immediacy). While this partially overlaps with Halliday’s interpersonal function of grammar, PGM treats these effects as dynamically arising within the process of speech generation, providing an extended supplement to functional grammar.

Thus, PGM has the potential to reconceptualize the correspondence between syntax and function not as a static system but as a variable and selective phenomenon within the process of generation, thereby connecting the network of social meanings depicted by SFL to a more actionable level of grammatical description.

D Grammar and Performance: The Structure of Narration and Immediate Grammar

In traditional linguistics, the perspective that the “performance effect” of narration is directly supported by grammatical structure has not been systematically addressed. Generative grammar has focused on meaning and structure, while pragmatics has concentrated on the speaker’s intention. However, there has been only partial mention of how the choice of syntax in narration can generate immediate performance effects (surprise, reveal, buildup).

In the Process Grammar Model (PGM), grammar is not merely a framework for meaning construction but functions as a “structure of performance” that allows the narrator to control the listener’s attention, emotions, and predictions. In particular, in immediate grammar, when the narrator presents information in the order they think of it on the spot, structures such as subject delay, preposing of locative phrases, and insertion of action phrases often occur. These structures have a performative effect of “setting the stage → drum-roll → focus disclosure” in cognitive processing, and the syntax itself supports that effect.

For example, the narration “From the mountains, here come the sheep!” follows the order “locative → verb phrase → subject,” structurally matching English Locative Inversion (From the box came a bird.). However, this word order is part of an immediately generated narrative flow and is not an effect of immediate use of adjusted expressions; it is an intrinsic syntax of immediate grammar.

In such examples, the performance does not depend on grammar; rather, grammar itself becomes the framework that enables the performance. Therefore, in PGM, the relationship between “grammar and performance” needs to clarify that the motivation for syntactic choice lies not only in “meaning” but also in “effect.”

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